Homework Assignment 4

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Chapter 5 (Question 1):

Suppose we have $x'(t) = v(x(t), t)$ and $x(0) = 0$ with

$$v(x, t) = 2t(1 + x).$$

Notice this is a separable equation.

$$\begin{aligned} x'(t) &= 2t(1+x) \\ \int \frac{1}{1+x}\, dx &= \int 2t\, dt \\ \ln|1+x| &= t^2 + C \\ 1 + x &= Ce^{t^2}. \end{aligned}$$

Substituting initial conditions, we have

$$\begin{aligned} 1 + 0 &= Ce^0 \\ C &= 0. \end{aligned}$$

Then substituting the constant back, we have

$$\begin{aligned} 1 + x &= e^{t^2} \\ x &= e^{t^2} - 1. \end{aligned}$$

Therefore, our solution is $x(t) = e^{t^2} - 1$.

Next, we compute the next 4 picard iterations denoted by X_n, starting with $X_0 = 0$. The first iteration:

$$\begin{aligned} X_1 &= X_0 + \int_0^t 2s(1 + X_0(s))\, ds \\ &= \int_0^t 2s\, ds \\ &= t^2. \end{aligned}$$

The second iteration:

$$\begin{aligned} X_2 &= \int_0^t 2s(1 + X_1(s))\ ds \\ &= 2\int_0^t s + s^3\ ds \\ &= 2(\frac{t^2}{2} + \frac{t^4}{4}) \\ &= t^2 + \frac{t^4}{2} \end{aligned}$$

The third iteration:

$$\begin{aligned} X_3 &= \int_0^t 2s(1 + X_2(s))\ ds \\ &= \int_0^t 2s(1 + s^2 + \frac{s^4}{2})\ ds \\ &= \int_0^t 2s + 2s^3 + s^5\ ds \\ &= t^2 + \frac{1}{2}t^4 + \frac{1}{6}t^6 \end{aligned}$$

The fourth iteration:

$$\begin{aligned} X_4 &= \int_0^t 2s(1 + X_3(s))\ ds \\ &= \int_0^t 2s(1 + s^2 + \frac{1}{2}s^4 + \frac{1}{6}s^6)\ ds \\ &= \int_0^t 2s + 2s^3 + s^5 + \frac{1}{3}s^7\ ds \\ &= t^2 + \frac{1}{2}t^4 + \frac{1}{6}t^6 + \frac{1}{24}t^8 \end{aligned}$$

Notice that this is the power series for e^{x^2} without the first term 1. Therefore, as $n \to \infty$, the iteration converges into the exact solution $e^{t^2} - 1$.

Chapter 5 (Question 2):

Suppose we have $\mathbf{x}'(t) = A(t)\mathbf{x}(t)$ with

$$A(t) = \begin{bmatrix} 0 & 1 \\ t^{-2} & -t^{-1} \end{bmatrix}.$$

Then, the flow transformation matrix is

$$[\Phi_{t,s}] = M(t)M^{-1}(s) = \frac{1}{2}\begin{bmatrix} \frac{t^2+s^2}{st} & \frac{t^2-s^2}{t} \\ \frac{t^2-s^2}{st^2} & \frac{t^2+s^2}{t^2} \end{bmatrix}.$$

Consider the taylor expansion of each term.
Top Left:

$$\frac{1}{2}\left(\frac{t^2+s^2}{st}\right) = 1 + \frac{1}{2s^2}(t-s)^2$$

Top Right:

$$\frac{1}{2}\left(\frac{t^2-s^2}{t}\right) = 0 + (t-s) - \frac{1}{2s}(t-s)^2$$

Bottom Left:

$$\frac{1}{2}\left(\frac{t^2-s^2}{st^2}\right) = 0 + \frac{1}{s^2}(t-s) - \frac{3}{2s^3}(t-s)^2$$

Bottom Right:

$$\frac{1}{2}\left(\frac{t^2}{s^2}\right) = 1 - \frac{1}{s}(t-s) + \frac{3}{2s^2}(t-s)^2$$

Then the taylor expansion the whole matrix looks like

$$[\Phi_{t,s}] = I + \begin{bmatrix} 0 & 1 \\ s^{-2} & s^{-1} \end{bmatrix}(t-s) + \frac{1}{2}\begin{bmatrix} s^{-2} & -s^{-1} \\ -3s^{-3} & 3s^{-2} \end{bmatrix}(t-s)^2$$

Starting from $\mathbf{X}_0 = \mathbf{x}_0$, we compute the next 3 terms of the Picard iteration, namely $\mathbf{X}_1, \mathbf{X}_2,$ and $\mathbf{X}_3$ for the solution of $\mathbf{x}'(t) = A(t)\mathbf{x}(t)$ and $\mathbf{x}(s) = \mathbf{x}_0$.

First, we compute the time-dependent matrices

$$I_A^{(k)}(t) = \int_s^t A(r) I_A^{(k-1)}(r) dr, \text{ for } k \in \{0,1,2,3\}.$$

This is because for all k, we have

$$\mathbf{X}_k(t) = \left(\sum_{j=0}^{k} I_A^{(j)}(t)\right)\mathbf{x}_0.$$

Notice that $I_A^{(0)} = I$. Then, we have

$$\begin{aligned} I_A^{(1)} &= \int_s^t A(r)\ dr \\ &= \begin{bmatrix} 0 & r \\ -\frac{1}{r} & -\ln|r| \end{bmatrix}_s^t \\ &= \begin{bmatrix} 0 & t-s \\ \frac{t-s}{st} & -\ln|\frac{t}{s}| \end{bmatrix}. \end{aligned}$$

This gives us our first iteration

$$\begin{aligned}\mathbf{X}_1 &= (I_A^{(0)} + I_A^{(1)})\mathbf{x}_0 \\ &= \begin{bmatrix} 1 & t-s \\ \frac{t-s}{st} & 1 - \ln\left|\frac{t}{s}\right| \end{bmatrix} \mathbf{x}_0.\end{aligned}$$

Notice how it matches the first order linear terms of the exact flow matrix's taylor expansion. Next, we have

$$\begin{aligned}I_A^{(2)} &= \int_s^t A(r) I_A^{(1)}(r)\, dr \\ &= \int_s^t \begin{bmatrix} \frac{1}{s} - \frac{1}{r} & -\ln\left|\frac{r}{s}\right| \\ r^{-2} - \frac{1}{s} r^{-1} & \frac{1}{r} - \frac{s}{r^2} + \frac{1}{r}\ln\left|\frac{r}{s}\right| \end{bmatrix} dr \\ &= \begin{bmatrix} \frac{r}{s} - \ln|r| & r - r\ln\left|\frac{r}{s}\right| \\ -r^{-1} - \frac{1}{s}\ln|r| & \ln|r| + sr^{-1} + \frac{1}{2}\ln^2\left|\frac{r}{s}\right| \end{bmatrix}_s^t \\ &= \begin{bmatrix} \frac{t}{s} - \ln\left|\frac{t}{s}\right| - 1 & t - t\ln\left|\frac{t}{s}\right| - s \\ s^{-1} - s^{-1}\ln\left|\frac{t}{s}\right| - t^{-1} & \frac{1}{2}\ln^2\left|\frac{t}{s}\right| + \ln\left|\frac{t}{s}\right| + st^{-1} - 1 \end{bmatrix}.\end{aligned}$$

Thus, our second iteration is

$$\begin{aligned}\mathbf{X}_2 &= (I_A^{(0)} + I_A^{(1)} + I_A^{(2)})\mathbf{x}_0 \\ &= \begin{bmatrix} ts^{-1} - \ln\left|\frac{t}{s}\right| & 2t - 2s - t\ln\left|\frac{t}{s}\right| \\ 2s^{-1} - 2t^{-1} - s^{-1}\ln\left|\frac{t}{s}\right| & \frac{1}{2}\ln^2\left|\frac{t}{s}\right| + st^{-1} \end{bmatrix} \mathbf{x}_0.\end{aligned}$$

Finally, we have

$$\begin{aligned}I_A^{(3)} &= \int_s^t A(r) I_A^{(2)}(r)\, dr \\ &= \begin{bmatrix} 2rs^{-1} - rs^{-1}\ln\left|\frac{r}{s}\right| - \ln|r| & \frac{1}{2}r\ln^2\left|\frac{r}{s}\right| + s\ln|r| - r \\ \frac{1}{2}s^{-1}\ln^2\left|\frac{r}{s}\right| + r^{-1}\ln\left|\frac{r}{s}\right| + r^{-1} & 2\ln|r| - \ln^2\left|\frac{r}{s}\right| + 2sr^{-1} - \frac{1}{6}\ln^3\left|\frac{r}{s}\right| \end{bmatrix}_s^t \\ &= \begin{bmatrix} 2ts^{-1} - ts^{-1}\ln\left|\frac{t}{s}\right| - \ln\left|\frac{t}{s}\right| - 2 & \frac{1}{2}t\ln^2\left|\frac{t}{s}\right| + s\ln\left|\frac{t}{s}\right| - t + s \\ \frac{1}{2}s^{-1}\ln^2\left|\frac{t}{s}\right| + t^{-1}\ln\left|\frac{t}{s}\right| + t^{-1} - s^{-1} & 2\ln\left|\frac{t}{s}\right| + 2st^{-1} - \ln^2\left|\frac{t}{s}\right| - \frac{1}{6}\ln^3\left|\frac{t}{s}\right| - 2 \end{bmatrix}.\end{aligned}$$

Thus, our third iteration is

$$\begin{aligned}\mathbf{X}_3 &= (I_A^{(0)} + I_A^{(1)} + I_A^{(2)} + I_A^{(3)})\mathbf{x}_0 \\ &= \begin{bmatrix} 3ts^{-1} - 2 - (2 + ts^{-1})\ln\left|\frac{t}{s}\right| & (t-s)(1 - \ln\left|\frac{t}{s}\right|) + \frac{1}{2}t\ln^2\left|\frac{t}{s}\right| \\ \frac{1}{2}s^{-1}\ln^2\left|\frac{t}{s}\right| + (t^{-1} - s^{-1})\ln\left|\frac{t}{s}\right| + s^{-1} - t^{-1} & -\frac{1}{6}\ln^3\left|\frac{t}{s}\right| - \frac{1}{2}\ln^2\left|\frac{t}{s}\right| + 2\ln\left|\frac{t}{s}\right| + 3st^{-1} - 2 \end{bmatrix} \mathbf{x}_0.\end{aligned}$$

Notice how each successive iteration gains higher order terms of the taylor series. Thus, by induction, you can see that the Picard iterations will eventually converge into the flow transformation.

Chapter 5 (Question 3):

Suppose we have the equation

$$t^2x''(t) - 2x(t) = \frac{3}{t} \text{ on } (0, \infty)$$

that satisfies $x(1) = 1$ and $x'(1) = 2$.

To find the solution, we first consider the associated homogenous equation:

$$t^2x''(t) - 2x(t) = 0.$$

We guess the solution $x(t) = t^r \mid r \in \mathbb{R}$. Then, we have

$$\begin{aligned} x'(t) &= rt^{r-1} \\ x''(t) &= r(r-1)t^{r-2}. \end{aligned}$$

Substituting our guessed solution, we have

$$\begin{aligned} t^2r(r-1)t^{r-2} - 2t^r &= (r(r-1) - 2)t^r \\ &= (r^2 - r - 2)t^r \\ &= ((r-2)(r+1))t^r \\ &= 0. \end{aligned}$$

Since $t > 0$, we divide through by t^r and get

$$r = 2 \text{ and } r = -1.$$

Thus, our homogenous solution is

$$x_h(t) = c_1t^2 + c_2t^{-1}.$$

Solving for the Wronskian, we have

$$\begin{aligned} W(x_1, x_2) &= det \begin{pmatrix} t^2 & t^{-1} \\ 2t & -t^{-2} \end{pmatrix} \\ &= -3. \end{aligned}$$

Now, we move on to the particular solution. Dividing the initial equation by t^2, we have

$$x''(t) - 2t^{-2}x(t) = 3t^{-3}.$$

Thus, $f(t) = 3t^{-3}$. Applying the Variation of Parameters formulas, we have

$$\begin{aligned} v_1 &= \int -\frac{(-t^{-1})(3t^{-3})}{-3}dt = \int t^{-4}dt = -\frac{1}{3}t^{-3} \\ v_2 &= \int \frac{t^2(3t^{-3})}{-3}dt = -\int t^{-1} = -\ln|t| \end{aligned}$$

Since v_1 is a constant multiple of the homogenous solution, it merges with C_2. Thus, our particular solution is

$$x_p(t) = -t^{-1} \ln |t|.$$

Combining the two solution gives us

$$\begin{aligned} x(t) &= C_1 t^2 + t^{-1}(C_2 - \ln |t|) \\ x'(t) &= 2C_1 t - t^{-2} C_2 - \frac{1 - \ln |t|}{t^2} \end{aligned}$$

Plugging in the initial condition $x(1) = 1$, we have

$$1 = C_1 + C_2. \tag{1}$$

Then, plugging in $x'(1) = 2$ gives us

$$3 = 2C_1 - C_2. \tag{2}$$

Adding the two equations and solving, we get

$$C_1 = \frac{4}{3} \text{ and } C_2 = -\frac{1}{3}.$$

Therefore, our final solution is

$$x(t) = \frac{4}{3} t^2 - t^{-1}(\frac{1}{3} + \ln |t|).$$

Chapter 5 (Question 7):

Suppose we have the equation

$$x''(t) - x'(t) - 6x(t) = e^t.$$

To find the general solution, we first consider the associated homogenous equation:

$$x''(t) - x'(t) - 6x(t) = 0.$$

We guess the following solution

$$\begin{aligned} x(t) &= e^{rt} \\ x'(t) &= re^{rt} \\ x''(t) &= r^2 e^{rt} \end{aligned}$$

Substituting, we get

$$\begin{aligned} r^2 e^{rt} - re^{rt} - 6e^{rt} &= e^{rt}(r^2 - r - 6) \\ &= r^2 - r - 6 \\ &= (r - 3)(r + 2) \\ &= 0. \end{aligned}$$

Thus, our homogenous solution is

$$x_h(t) = C_1e^{3t} + C_2e^{-2t}.$$

Solving for the Wronskian, we have

$$\begin{aligned} W(x_1, x_2) &= det \begin{pmatrix} e^{3t} & e^{-2t} \\ 3e^{3t} & -2e^{-2t} \end{pmatrix} \\ &= -5e^t. \end{aligned}$$

Moving to the particular solution. Notice that $f(t) = e^t$. Then, applying the Variation of Parameters, we have

$$\begin{aligned} v_1 &= -\int \frac{e^{-2t}e^t}{-5e^t} = \frac{1}{5}\int e^{-2t} = -\frac{1}{10}e^{-2t} \\ v_2 &= -\frac{1}{5}\int e^{3t} = -\frac{1}{15}e^{3t} \end{aligned}$$

Thus, our particular solution is

$$\begin{aligned} x_p(t) &= -\frac{1}{10}e^{-2t}e^{3t} - \frac{1}{15}e^{3t}e^{-2t} \\ &= -\frac{3}{30}e^t - \frac{2}{30}e^t \\ &= -\frac{1}{6}e^t. \end{aligned}$$

Therefore, the general solution is

$$x(t) = C_1e^{3t} + C_2e^{-2t} - \frac{1}{6}e^t.$$